

Task-6.

Implement various text file operation.

Aim: To write a python program implement various text file operations.

Problem 6.1: you need to write the sentence "Error objects are thrown when runtime errors occur. The Error object can also be used as a base object can also used as a base object for user-defined exceptions" into a text file named log.txt. Implement a function that performs this task.

Algorithm:

1. write to a file:

- Define writefile(filename) function:

- open a file named "log.txt" in write mode.
- write the following text to the file.

- "Error objects are thrown when runtime errors occur. The Error object can also be used as a base object for user-defined exceptions".

- close the file.

2. Read from a file.

- Define readfile(filename) function:

- open the file specified by filename in read mode using a with statement.

- Read the entire content of the file

- print the content

3. Execute the program:

- Call writefile("write") to write the predefined text to "log.txt"

- Call readfile("text") to attempt to read from a file named "text" and print its content

program 6.1

```
def writefile(filename):
```

```
    f=open("log.txt","w")
```

```
    f.write("Error objects are thrown when runtime errors  
    occur. The Error object can also be used as a base  
    object for user-defined exceptions")
```

```
    f.close()
```

```
def readfile(filename):
```

```
    with open(filename, "r") as file:
```

```
        content=file.read()
```

```
        print(content)
```

```
writefile("write")
```

```
readfile("text")
```



Output:

error objects are thrown when runtime errors occur.

The error object can also be used as a base object for user-defined exceptions.

- Define writefile(filename) function:
 - open a file named "log.txt" in write mode.
 - write the following text to the file.
 - Error object are thrown when runtime error occur. The error object can also be used as base object for user-defined exceptions.
 - Close the file.
- Read from a file:
 - Define readfile(filename) function.
 - open the file specified in filename in read mode.
 - Read the entire content of the file.
 - Print the content.
- Execute the program:
 - Call writefile("log.txt") to write the predefined text to "log.txt".
 - Call readfile("log.txt") to attempt to read from "log.txt" and print its content.

prog
def w
f=
f=
o
o
f
def x

writ
read

program 6.2

```
def count_error_lines(filename):
```

```
    error_count = 0
```

```
    with open(filename, "r") as file:
```

```
        for line in file:
```

```
            if "ERROR" in line:
```

```
                error_count += 1
```

```
    return error_count
```

```
error_lines = count_error_lines("log.txt")
```

```
print(f"Number of lines with 'ERROR': {error_lines}")
```

log.txt

"Error objects are thrown when runtime error occurs
The Error object can also be used as a base object for
user-defined exceptions."

problem 6.3:

You need to write a report containing the details (Name, departments) of the employee in list. write a Python function that writes this report to a file named `employee-report.txt`

Algorithm:

1. create Employee Data:
 - Define the function `write-employee-report(filename)`
 - create a list `employees` containing dictionaries, each with "name" and "department" keys for individual employees.
2. open file for writing:
 - open the file specified by `filename` in write mode using a `with` statement.
3. write Employee Data to file:
 - Loop through each employee in the `employees` list
 - for each employee, format a string as "Name: {employee['name']}, Department: {employee['department']}"
 - write the formatted string to the file, followed by a new line character (`\n`).
4. Execute the program.
 - call `write-employee-report("employee-report.txt")` to write the employee data to the file "employee-report.txt".

Program 6.3.

```
def write_employee_report(filename):
```

```
    employees = [
```

```
        {"name": "Alice", "department": "HR"},
```

```
        {"name": "Bob", "department": "Engineering"},
```

```
        {"name": "Charlie", "department": "Finance"}]
```

```
    with open(filename, "w") as file:
```

```
        for employee in employees:
```

```
            line = f"Name: {employee['name']}, Department:
```

```
                    {employee['department']} \n"
            file.write(line)
```

Example usage:

```
write_employee_report("employee_report.txt")
```

VEL TECH	
EX No	6
PERFORMANCE (5)	5
ANALYSIS (5)	5
VOICE (5)	5
RECORD (5)	5
STATUS	15
SIGN WITH DATE	17/5

Result:

Thus the Python program Implement various text-file operations was successfully executed and the output was verified.